ATVED: Installation & Run Guide

1. Prerequisites & Installation

Before running ATVED, ensure you have Python 3.10 or higher installed. For maximum performance, it is highly recommended to have an NVIDIA GPU and install the CUDA version of PyTorch.

Step 1: Install base dependencies:
pip install fastapi uvicorn sqlalchemy aiosqlite fpdf ultralytics opencv-python easyocr requests

Step 2 (Optional but recommended): Install GPU-accelerated PyTorch:
pip uninstall torch torchvision torchaudio -y
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118

2. Starting the Backend Database & API

The backend server acts as the brain of the operation, recording violations and generating E-Challan PDFs.

Open a terminal in the ATVED_Gridlock folder and run:
python api_server.py

Leave this terminal open. Your backend is now securely running on http://localhost:8000.

3. Starting the AI Vision Engine

With the backend running, you can now start the computer vision pipeline to process traffic video.

Open a SECOND terminal window in the ATVED_Gridlock folder and run:
python computer_vision_models/demo_live_end_to_end.py traffic.mp4

(Note: To use a live webcam instead of the provided video, replace 'traffic.mp4' with '0')

The engine will load the YOLO and OCR models onto your GPU and begin processing the video, pushing violations to the backend in real-time.

4. Viewing the Dashboards

Now that both the backend and AI engine are running, you can monitor the system via the Web Dashboards.

If you are running everything locally, open your web browser to:

- Police Control Room: http://localhost:8000/authority
- Citizen Portal: http://localhost:8000/user

If you have deployed your dashboards to Netlify, simply visit your Netlify URL (e.g., https://your-site.netlify.app). The Netlify dashboard will automatically securely fetch data from your local backend!